CWL workflows with MPI in bare-metal, containers, cloud, and HPC

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July 6, 2026

Agenda

- **Introduction**
- **Motivation**
- **Methodology**
- **Results**
- **Conclusion**

Introduction

Introduction — MPI and CWL

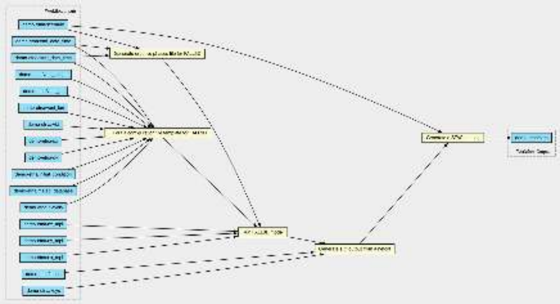
MPI

COMMON WORKFLOW LANGUAGE

A **portable** message-passing **standard** for **parallel programming** widely used in HPC.

A **standard** for describing **computational workflows**, focused on **portability** and reproducibility across environments.

Introduction — Computational Workflows

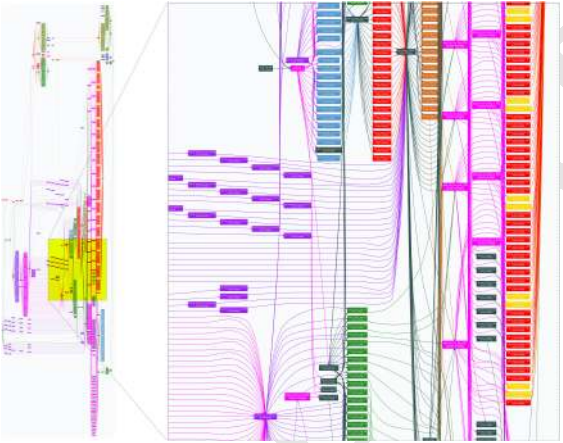


Source
GEO3BCN-CSIC
FALL3D Workflow

GET-IT workflows for Etna and La Palma test cases

https://gitlab.geo3bcn.csic.es/fall3d/getit-workflows

Introduction — Computational Workflows

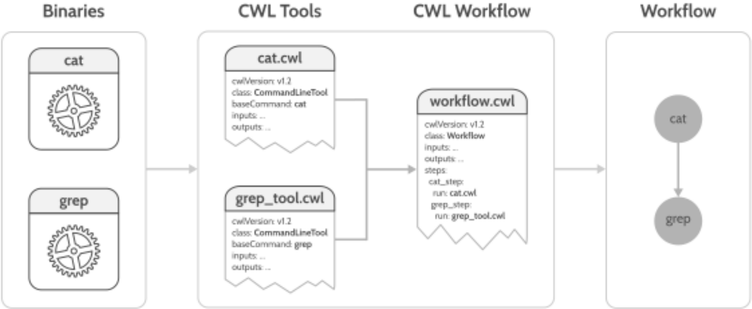


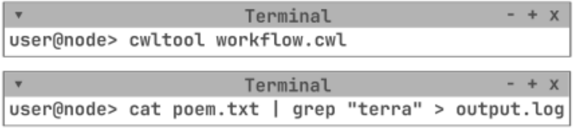
Source
H. Oliver et al. (2019)
Workflow Automation for Cycling Systems

Unified Model (UM) example with Cylc.

Computing in Science & Engineering (IEEE)
DOI: 10.1109/MCSE.2019.2906593

Introduction — Common Workflow Language (CWL)





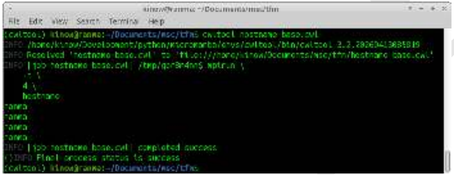
Introduction — MPIRequirement

```
$ cwltool hostname_base.cwl
```

```yaml
# File: hostname_base.cwl
cwlVersion: v1.2
class: CommandLineTool

baseCommand: mpirun
arguments:
  - "-np"
  - "4"
  - "hostname"

inputs: []
outputs: []
```



```
$ cwltool --mpi-config-file mpi-config.cwl \
    hostname_mpireq.cwl
```

```yaml
# File: hostname_mpireq.cwl
cwlVersion: v1.2
class: CommandLineTool

requirements:
  - class: "cwltool:MPIRequirement"
    processes: 4

baseCommand: hostname

inputs: []
outputs: []
```

MPI configuration file

```yaml
# File: mpi-config.cwl
runner: srun
nproc_flag: "-n"
extra_flags: "--hostfile /shared/hostfile"
```

Motivation

Motivation

The portability of MPI-enabled workflows in CWL remains largely unvalidated.

The `MPIRequirement` extension (Nash et al., 2020) has not been systematically evaluated across CWL runners and execution environments.

Nash et al. (2020), WORKS 2020 (IEEE/ACM)
DOI: 10.1109/WORKS51914.2020.00008

Methodology

Methodology

Experiment	Variables	Values
CWL Conformance Tests	Runner	cwltool, Toil, StreamFlow
	HPC System	MN5, FT3, LUMI
	Execution Mode	login-node, slurm-node, slurm-batch
MPI Workflow Evaluation	Runner	cwltool, Toil
	Environment	HPC, Cloud, Bare-metal
	MPI Integration	Launcher-based, `MPIRequirement`
	Containerisation	Enabled, Disabled

Results

Results — CWL Conformance Tests

CWL Conformance Tests

Results — CWL Conformance Tests

cwltool

v	FT3		MN5		LUMI	
	l	s	l	s	l	s
1.0.2						
1.1.0						
1.2.1						

StreamFlow

v	FT3		MN5		LUMI	
	l	s	l	s	l	s
1.0.2						
1.1.0						
1.2.1						

Toil

v	FT3			MN5			LUMI		
	l	s	b	l	s	b	l	s	b
1.0.2									
1.1.0									
1.2.1									

≥ 95%
70–95%
< 70%
not run

l login-node
s slurm-node
b slurm-batch

Results — CWL Conformance Tests

StreamFlow

v	FT3		MN5		LUMI	
	l	s	l	s	l	s
1.0.2	∘	∘	∘	△∘		
1.1.0						
1.2.1			∘	△∘		

Toil

v	FT3			MN5			LUMI		
	l	s	b	l	s	b	l	s	b
1.0.2			△□		△	△□			△□
1.1.0			△□		△	△□			△□
1.2.1			△□		△	△□			△□

Observed Issues

- △ No network access
- ∘ Singularity container issues
- □ Slurm submission errors
- Symlink issues
- Race conditions
- Tests bootstrap problems in HPC
- Slurm compute node /tmp
- Intermittent failures

- 13 issues reported already

Results — MPI Workflows

Simple MPI Workflow

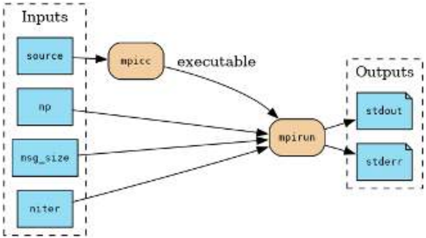
Results — Simple MPI Workflow

baseCommand: mpirun

Mode	Laptop	Cloud	MN5	LUMI
cwltool (n)				
cwltool (s)				
Toil (n)				
Toil (s)				

MPIRequirement

Mode	Laptop	Cloud	MN5	LUMI
cwltool (n)				
cwltool (s)				
Toil (n)				
Toil (s)				



OK
ERROR
INVALID*

(n) No container
(s) Singularity

* MPI launched through the container rather than the host launcher.

Results — Simple MPI Workflow

baseCommand: mpirun

Mode	Laptop	Cloud	MN5	LUMI
cwltool (n)				□
cwltool (s)			△	
Toil (n)			△	□
Toil (s)			△	

MPIRequirement

Mode	Laptop	Cloud	MN5	LUMI
cwltool (n)				
cwltool (s)	∘	∘	∘	∘
Toil (n)			△	
Toil (s)	∘	∘	∘	∘

Observed Issues

- △ No network access
- ∘ cwltool `MPIRequirement` bug
- □ No mpirun
- `singularity mpirun...`
- 1 issue + 1 pull request in cwltool

OK
ERROR
INVALID*

(n) No container
(s) Singularity

* MPI launched through the container rather than the host launcher.

Results — MPI Workflows

FALL3D Workflow

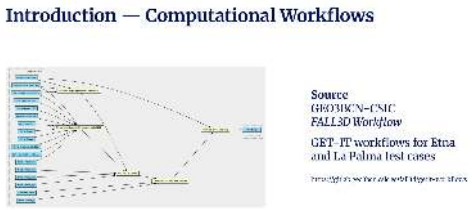
Results — FALL3D Workflow

baseCommand: mpirun

Mode	Laptop	Cloud	LUMI
cwltool (n)			
cwltool (s)			

MPIRequirement

Mode	Laptop	Cloud	LUMI
cwltool (n)			
cwltool (s)			



OK
ERROR
INVALID*

(n) No container
(s) Singularity

* MPI launched through the container rather than the host launcher.

Results — FALL3D Workflow

baseCommand: mpirun

Mode	Laptop	Cloud	LUMI
cwltool (n)	△		□
cwltool (s)		■	

MPIRequirement

Mode	Laptop	Cloud	LUMI
cwltool (n)			
cwltool (s)	∘	∘	∘

Observed Issues

- △ ABI-compatibility issue
- □ no mpirun
- ■ MPI oversubscription
- ∘ cwltool `MPIRequirement` bug
- `singularity mpirun...`
- 2 issues + 1 pull request in cwltool

OK
ERROR
INVALID*

(n)	**No container**
(s)	**Singularity**

* MPI launched through the container rather than the host launcher.

Conclusion

Conclusion — Objectives

- Evaluate the capability of existing CWL runners to execute MPI-based workflows on HPC systems and cloud platforms.
- Analyse the behaviour and portability of the `MPIRequirement` CWL extension with `cwltool` across different execution environments.
- Compare the compatibility and limitations of different CWL runners across execution environments.
- Assess MPI execution using both simple illustrative workflows and a representative large-scale scientific application.
- Characterise the current state of MPI support in CWL-based systems across bare-metal, cloud, and HPC environments.

Conclusion — Key Findings

Main insights

- **`MPIRequirement` improves portability and execution success across heterogeneous environments.**
- CWL runners differ in HPC compatibility, especially for MPI and Slurm execution.

Limitations

- Container and runtime integration issues still limit full portability in some HPC systems.
- `MPIRequirement` is not yet part of the CWL standards.

Open Source Contributions

- A total of 32 issues related to CWL, MPI, & HPC reported to 12 OSS projects.

Conclusion — Future work

MPIRequirement

- Improve HPC scheduler integration (Slurm, PBS, LSF, Flux, ...) in CWL runners.
- Strengthen MPI + container interoperability in CWL.
- Inclusion of `MPIRequirement` in CWL standards and additional CWL runners.

Broader validation

- Extend evaluation to more CWL runners and HPC systems.
- Evaluate larger MPI-based scientific workflows.
- Address issues identified in evaluated open-source CWL runners.

Beyond MPI

- Support heterogeneous paradigms (MPI+X, OpenMP, CUDA, HIP, ...).

Data Availability

All experimental artefacts are publicly available.

`https://doi.org/10.5281/zenodo.20348637`

`https://github.com/kinow/cwl-mpi`

RO-Crate metadata included for reproducibility and reuse.

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Questions?

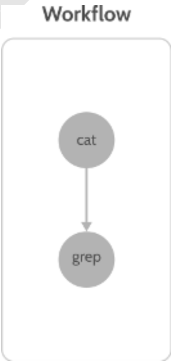
Thank you.

Introduction — Message Passing Interface (MPI)

- Standard API for parallel programming.
  - MPICH, Open MPI, Intel MPI, Cray MPI, Fujitsu MPI, RIKEN-MPICH, …
- Executes multiple processes that communicate by passing messages.
- Typically launched using `mpirun`, `mpiexec`, or `srun`.
- Commonly used in **scientific and HPC applications**.

Introduction — Computational Workflows

- A computational workflow is a collection of tasks with dependencies.
- Dependencies are represented as a graph (Directed Acyclic Graph [DAG] or Directed Cyclic Graph [DCG]).
- Workflow Management Systems (WfMS) execute tasks while respecting these dependencies.

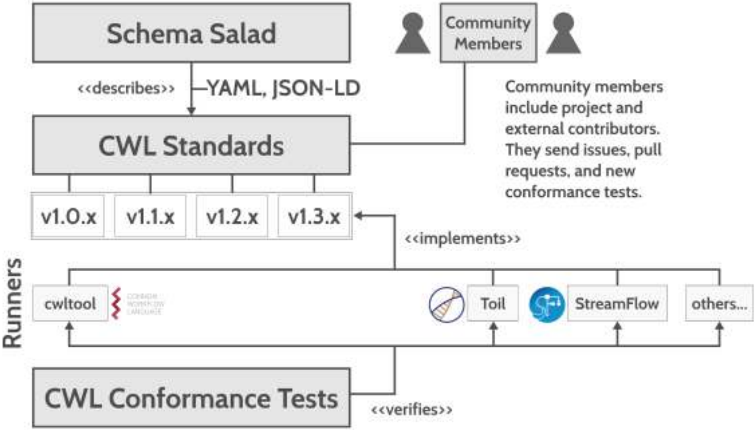


Introduction — Common Workflow Language (CWL)

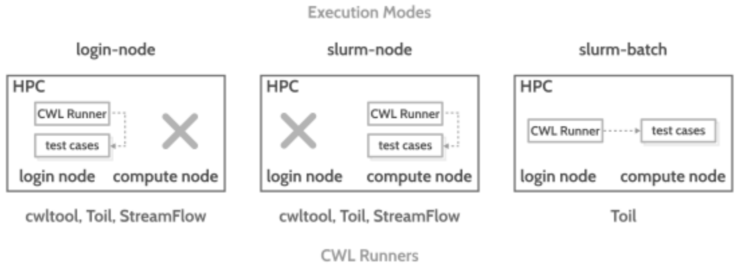
- Open standards for describing computational workflows.
  - cwltool, Toil, StreamFlow, Arvados, SciWin, Ravanan, Calrissian, REANA, Pegasus, Galaxy, CWL-Airflow, …
- Promotes portability and reproducibility across platforms.
- Workflows consist of `CommandLineTools` connected through workflow steps.
- The standards provide extensions as requirements & hints.
  - `DockerRequirement`, `SoftwareRequirement`, `NetworkAccess`, MPIRequirement, …

COMMON WORKFLOW LANGUAGE

Introduction — Common Workflow Language (CWL)



Methodology — Execution modes



Performance — MPI Bandwidth Comparison (`osu_bw`)

LUMI MPI point-to-point bandwidth benchmark (`pt2pt/osu_bw`, `--mpi=pmi2`).

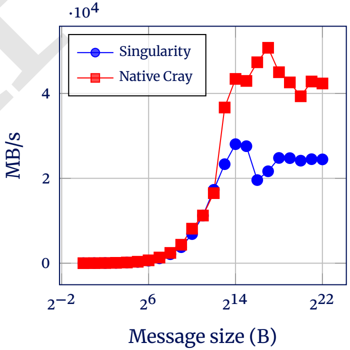
Singularity

Size	BW
1	10.25
2	20.01
8	81.94
32	299.38
128	1153.15
512	3777.49
2048	11220.21
8192	23334.12
16384	28045.90
65536	19587.88
131072	21663.76
524288	24758.18
2097152	24510.72
4194304	24461.10

Native Cray

Size	BW
1	8.53
2	21.18
8	86.06
32	344.33
128	1340.01
512	4353.42
2048	11235.76
8192	36684.87
16384	43421.54
65536	47344.88
131072	50755.53
524288	42599.48
2097152	42816.48
4194304	42347.43

Comparison



FALL3D — Output